Yiguang Zhu

ORISE Fellow, FDA/CDER

Rockville, MD | (814) 996-9046 | yiguang.zhu@fda.hhs.gov | Personal Website | LinkedIn

Current Appointment

ORISE Fellow | U.S. Food and Drug Administration — White Oak, MD
Office of New Drug, Center for Drug Evaluation and Research — Oct. 2026 – Present

EDUCATION

Johns Hopkins University | Whiting School of Engineering — Baltimore, MD
PhD in Environmental Health and Engineering — Jul. 2021 – Oct. 2026

- Advisor: Dr. Paul A. Locke, JHU Toxicology Policy Program
- Research area: Validation and regulatory acceptance of New Approach Methodologies (NAMs), with a focus on cardiac in vitro models for drug-induced cardiotoxicity screening

Johns Hopkins University | Bloomberg School of Public Health — Baltimore, MD
ScM in Environmental Health and Engineering - GPA: 3.95 — Aug. 2019 – May 2021

- Advisor: Dr. Paul A. Locke
- Thesis: *Reducing Lung Cancer Risk from Radon Exposure in Manheim Township, PA: Radon Resistant New Construction (RRNC) as a Significant Public Health Tool*

Pennsylvania State University | College of Earth and Mineral Sciences — State College, PA
BS in Environmental Systems Engineering - GPA: 3.73 — Aug. 2015 – May 2019

Research Interests

My research focuses on the intersection of biomedical innovation and regulatory science. I study how **New Approach Methodologies (NAMs)**, especially complex in vitro models (CIVMs) such as organoids and organ-on-a-chip systems, can be validated and adopted for regulatory decision-making. My goal is to strengthen **evidence-based policymaking** and accelerate the adoption of innovative, human-relevant testing methods.

Skills

Data & Software: R, MATLAB, Stata, GraphPad Prism, ImageJ/Fiji, BioRender
Languages: Chinese, English
Regulatory & Communication: Data-driven policy analysis, scientific communication, stakeholder engagement, regulatory/technical writing
Wet-lab & in vitro: cell culture, hiPSC differentiation, tissue engineering, electrophysiology/functional assay readouts, immunostaining, adenovirus/AAV transduction

Research Experiences

Doctoral Research | Johns Hopkins University — Jul. 2021 – Oct. 2026
Advisor: Dr. Paul A. Locke

- Developed a regulatory validation framework to support the acceptance of NAMs in toxicity testing, drug development, and regulatory decision-making
- Applied this framework retrospectively to three hiPSC-based CIVM, CHAMP (2D cardiomyocyte monolayers and 3D cardiac organoids), STRIDE-96 (engineered heart tissues), and NanoMEA (nanopatterned microelectrode arrays), to evaluate regulatory applicability for drug-induced cardiotoxicity screening

Visiting Student | Johns Hopkins University — Aug. 2022 – Nov. 2024
Supervisor: Dr. Deok-Ho Kim, Department of Biomedical Engineering

- Optimized a nanopatterned microelectrode array (nanoMEA) based functional phenotyping pipeline for drug-induced cardiotoxicity screening (published)

- Developed and validated a high-throughput, automated Engineered Heart Tissue (EHT) model for physiological assessment and cardiotoxicity screening (manuscript submitted)

Doctoral Researcher | Johns Hopkins University May 2020 – Oct. 2026
Toxicology Policy Program
- Authored white papers advocating for laboratory animal welfare and non-animal testing methods; language was adopted in 4 congressional committee reports accompanying appropriations bills
- Facilitated numerous meetings with congressional offices and assisted in legislative hearings and briefings

ScM Thesis Research | Johns Hopkins University Aug. 2019 – May 2021
Advisor: Dr. Paul A. Locke
- Evaluated public health impacts of local radon mitigation ordinances using 1,734 radon tests from 483 homes, identified through iterative cleaning and screening of statewide Pennsylvania radon data (published)
- Synthesized epidemiological data into policy recommendations and presented in four stakeholder meetings

Undergraduate Independent Study | Pennsylvania State University Oct. 2018 – May 2019
Advisor: Dr. Meng Wang
- Conducted a literature review on integrative phototrophy for environmental and public health services
- Studied the production pathways of biohydrogen and polyhydroxyalkanoates (PHAs) by microalgae

Research/Lab Assistant | Fujian Normal University Summer 2017 and 2018
Supervisor: Dr. Yiqing Li, State Key Laboratory of Subtropical Mountain Ecology
- Participated in field-based soil warming experiments, including forest litter collection, biometric measurements, and photosynthetic parameter monitoring (published)

Employment History

Intern | National Center for Health Research, Washington, DC Jun. 2025 – Aug. 2025
- Tracked FDA and congressional activities (e.g., Advisory Committee meetings, House/Senate hearings)
- Analyzed FDA programs and pathways (user fees, 510(k)/DDT/PMA, EUA, accelerated approval); drafted a manuscript examining the benefits and limitations of COVID-19 EUAs and transition to full approval
- Contributed to FDA docket comments and policy statements (e.g., hormone replacement therapy), aligning recommendations with patient safety and evidence standards

Lead Teaching Assistant | Bloomberg School of Public Health Mar. 2022 – Present
- Managed the TA team for Introduction to Environmental and Occupational Health Law
- Moderated class discussions, addressed student inquiries, and graded students' essays

English-Chinese Translator/Interpreter | Freelance Feb. 2015 – Present
- Provided English–Chinese translation and interpretation for academic, business, legal, and scientific settings

Grading Assistant | Pennsylvania State University Jan. 2017 – May 2019
Civil and Environmental Engineering Department, Pennsylvania State University
- Worked with three instructors on the course Introduction to Environmental Engineering (class size: 80+)

Intern | The Amity Foundation, Nanjing, China Summer 2015
- Translated 100+ patient case files for low-income children with Down syndrome or cleft lip from Chinese to English, ensuring culturally sensitive communication with international donors
- Coordinated 2 volunteer outreach campaigns focused on raising awareness and funds for cleft lip surgeries and Down syndrome

Peer-Reviewed Journal Publications

Zhu, Y., & Locke, P. A. (2026). From application-first to science-first: A two-phase validation framework for complex in vitro NAMs. *Clinical Pharmacology & Therapeutics*, 120(5), 1173–1177. doi:10.1002/cpt.70428

Zhu, Y., Latshaw, M. W., & Locke, P. A. (2026). Bright spot, closing door: Encouraging innovation, restricting foreign talent [Editorial]. *American Journal of Public Health*, 116(4), 482–484. doi:10.2105/AJPH.2025.308384

Ren, Z., Kim, B., Go, G., Wang, H., Cheng, K., **Zhu, Y.**, Matsubara, T., Harriot, A.D., Loyala, A., Akarapipad, P., Criscione, J., Herron, T., Lee, P.H.U., Lee, D.I., Lee, L.P.S., Kim, D.-H. A High-Throughput Human Cardiac Muscle Chip Enabling Multimodal Functional Profiling and Machine Learning-Driven Cardiotoxicity Screening. *Nature Communications.* (under review)

Zhu, Y., Fox, M. A., Corrigan, B., & Locke, P. A. (2026). Reducing lung cancer risk from radon exposure in Manheim Township, PA: Radon Resistant New Construction (RRNC) as a significant public health tool. *International Journal of Radiation Biology*, 102(1), 38–48. doi:10.1080/09553002.2025.2588394

Kim, B., Choi, J. S., **Zhu, Y.**, Kim, J., Kim, Y. S., Parra, A., Locke, P. A., Kim, J. H., Herron, T., & Kim, D.-H. (2025). Effect of electrochemical topology on detection sensitivity in MEA assay for drug-induced cardiotoxicity screening. *Biosensors and Bioelectronics*, 272, 117082. doi:10.1016/j.bios.2024.117082

Ye, W., Xiong, D., Yang, Z., **Zhu, Y.**, Zhang, Q., Lin, W., Xu, C., Liu, X., Zhang, J., & Yang, Y. (2019). Effect of soil warming on growth and photosynthetic characteristics of *Cunninghamia lanceolata* saplings. *Acta Ecologica Sinica*. doi:10.5846/stxb201801150110

Presentations

Zhu, Y., Gabrielson, K., Kohr, M., Machairaki, V., Smirnova, L., & Locke, P. A. (2026). *Two-Phase Validation Framework Informed by an In Vitro Cardiotoxicity Screening Case Study*. Poster presentation at the Society of Toxicologic Pathology 45th Annual Symposium (STP 2026), San Diego, CA.

Zhu, Y., & Locke, P. A. (2026). *Advancing Regulatory Acceptance of NAMs: A Two-Phase Validation Framework Informed by Cardiac MPS Case Studies*. Poster presentation at the Microphysiological Systems World Summit 2026 (MPS World Summit 2026), Washington, DC.

Zhu, Y., & Locke, P. A. (2025). *Advancing Validation Methodologies and Regulatory Acceptance of Microphysiological Systems*. Poster Presentation, EHE Research Retreat Poster Session, Baltimore, MD.

Ren, Z., Kim, B., Go, G., Wang, H., **Zhu, Y.**, Akarapipad, P., Loyala, A., Matsubara, T., Criscione, J., Lee, D., Herron, T., & Kim, D. (2024). *Automated and Scalable Muscle Tissue System with Magnetic-based Functional Evaluation*. Presentation at the Biomedical Engineering Society 2024 Annual Meeting (BMES 2024), Baltimore, MD.

Zhu, Y., & Locke, P. A. (2024). *Advancing Validation Methodologies and Regulatory Acceptance of Microphysiological Systems*. Oral Presentation, European Society for Alternatives to Animal Testing (EUSAAT 2024), Linz, Austria.

Locke, P. A., & **Zhu, Y.** (2024). *Reducing Lung Cancer Risk from Radon Exposure: Radon Resistant New Construction as a Significant Public Health Tool*. Oral Presentation, Canadian Nuclear Laboratories' Low Dose Radiation Webinar Series, Virtual.

Zhu, Y. (2024). *The Impact of U.S. FDA Modernization Act 2.0 on Animal Testing*. Oral Presentation, Personal Care and Homecare Ingredients (PCHi 2024), Shanghai, China (Virtual).

Zhu, Y., & Locke, P. A. (2024). *Advancing Validation Methodologies and Regulatory Acceptance of Microphysiological Systems*. Poster Presentation, Microphysiological System World Summit 2024 (MPS WS 2024), Seattle, WA.

Zhu, Y., & Locke, P. A. (2023). *Reducing Lung Cancer Risk from Radon Exposure in Manheim Township, PA: Radon Resistant New Construction (RRNC) as a Significant Public Health Tool*. Oral Presentation, Indoor Environments 2023™, Nashville, TN.

Zhu, Y., & Locke, P. A. (2023). *Recommendations for Promoting the Regulatory Acceptance of Microphysiological Systems*. Oral Presentation, 12th World Congress on Alternatives and Animal Use in the Life Sciences (WC12), Niagara Falls, Canada.

Zhu, Y. (2021). *RRNC Cost Effectiveness for Reducing Lung Cancer Risk*. Oral Presentation, CRCPD - National Radon Training Event, Virtual.

Zhu, Y., & Locke, P. A. (2020). *Lung Cancer Risk Reduction Associated with RRNC: A study of Manheim Township, PA*. Presentation at the Region 3 Mid-Atlantic Radon Stakeholder Meeting, Virtual.

Zhu, Y. (2019). *Phototrophs Polyhydroxyalkanoates (PHAs) and Hydrogen Production from Wastewater*. Poster Presentation, Undergraduate Exhibition, State College, PA.

Leadership & Service

SPDG Science Policy Coordinator | Johns Hopkins University Jun. 2025 – Present
- Served on the leadership team at Science Policy & Diplomacy Group
- Spearheaded networking events and career panels, connecting graduate students with industry leaders

PhD Representative & GRO Liaison | Johns Hopkins University Aug. 2022 – Aug. 2024
- Represented over 50 PhD students at the Graduate Representative Organization (GRO)
- Initiated and organized volunteer and social events to cultivate supportive departmental community

CSSA Benefiting THON Chair | Pennsylvania State University Aug. 2017 – May 2019
- Organized campus-wide fundraisers with my team and raised over $6000 for pediatric cancer research
- Completed the 46-hour, no-sitting, no-sleeping dance marathon

Honors and Awards

Radon Mini-Grant Program | Conference of Radiation Control Program Directors 2020
Morgan-James Scholarship Fund | Bloomberg School of Public Health Fall 2020 & Fall 2024
John and Elizabeth Holmes Teas Scholarship | Pennsylvania State University Fall 2018
Dean's List | Pennsylvania State University 2015–2019

Certificates

Certificate of Risk Sciences and Public Policy | Johns Hopkins University 2021
Certificate of Food Systems and Public Health | Johns Hopkins University 2021
GIA Applied Jewelry Professional (GIA AJP) | Gemological Institute of America (GIA) 2020